Figure 4: conga-IT6 Mini-ITX motherboard (Credit: congatec)

range of software tools and programming environments for deep learning and AI applications are available. These include popular open source platform ROCm for GPGPU applications.

The open source idea is particularly important in this context, as it prevents OEMs from becoming dependent on a proprietary solution. HIPfy is an available tool with which proprietary applications can be transferred into portable HIP C++ applications, so that dangerous dependence on individual GPU manufacturers can be effectively avoided.

AI development has also become much easier with the availability of OpenCL 2.2, because since then, OpenCL C++ kernel language has been integrated into OpenCL, which makes writing parallel programs much easier. With such an ecosystem, both knowledge-based AI and deep learning are comparatively easy to implement, and are no longer just the reserve of billion-dollar IT giants like Google, Apple, Microsoft and Facebook.

Fast design-in with COMs

Now the only remaining question is: How can OEMs design these hardware-based AI enablers into their applications as quickly and efficiently as possible? One of the most efficient ways is via standardised computer-on-modules (COMs) that have been equipped with comprehensive support for GPGPU processing.

COMs have a slim footprint, support designs with OEM-specific feature sets, and as application-ready super components these come pre-integrated with everything that developers would otherwise have to arduously assemble in a full custom design. This makes COMs crucial for faster time-to-market.

As these are not only application-ready but also functionally-validated by many users, COMs offer high design security. This means that OEMs can expect to save between 50 and 90 per cent of their NRE (non-recurring engineering) costs when using modules. Thanks to the modular approach, the application can also be scaled to requirements. With a simple module swap, new performance classes can be integrated into existing carrier board designs without any further design effort, so that OEMs can easily add these innovative features to the functionality of their designs.

COM Express, the leading standard for high-end modules

A leading form factor among modules for this performance class is COM Express standard. This has been developed over many years by standards development organisation PICMG and is supported by all leading embedded computing suppliers. Companies like congatec—who have long been working closely with AMD and recently even extended the availability of AMD Geode processors—offer AMD Ryzen Embedded V1000 processor-based modules, for example, in COM Express Basic Type 6 form factor, which offers sufficient capacity to cover the entire performance range from 15 watts to 54 watts TDP.

Thanks to RTS hypervisor support, the real-time capable conga-TR4 module can also support AI platforms where deep learning systems and digital twins are to be connected via virtual machines so that hard real-time processing can always be ensured.

And with its standardised APIs, the module is also prepared for data exchange via the Internet of Things (IoT) gateways in compliance with SGET UIC standard. This allows OEMs to fully concentrate on developing the application—any missing glue logic can be specifically-developed and provided upon customer request.

Embedded high-end edge server design options

For OEMs wanting to bring their digital twins and deep learning intelligence to the industrial edge, there are many attractive options from companies like congatec. For example, embedded designs based on AMD EPYC Embedded 3000 processors come with long-term availability of up to 10 years. With up to 16 cores, 10-Gigabit Ethernet performance and up to 64 PCIe lanes, processors of the embedded server-class even enable deep learning applications at the edge of the Industrial IoT (IIoT). Hence, developers have everything they need on the hardware side for deep learning-based AI platforms for real-time industrial use. END

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Zeljko Loncaric

The author is working as a marketing engineer at congatec AG.